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Foundations of Blockchain Technology

Essay

The Synergy of Blockchain and Artificial Intelligence

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1 Introduction

Two technologies have shaped the digital world of the past decade — *blockchain* and *artificial intelligence* (AI). Artificial intelligence gave machines the ability to learn from data, recognise patterns, generate text and make decisions; blockchain gave us a way to store and transfer value and data without an intermediary, in an environment where the participants do not trust one another. For a long time they were treated separately, yet it is becoming increasingly clear that they are *complementary* technologies: the strengths of one line up precisely with the weaknesses of the other.

Artificial intelligence is extremely powerful, but it also has fundamental flaws. It is largely **centralised** (models are controlled by a handful of large corporations), it operates as a **black box** (it is hard to explain *why* a model made a particular decision), and its quality depends entirely on **trust in the training data**, which can be substituted or poisoned. Blockchain, by contrast, is **decentralised**, **transparent**, provides **immutability** of records and verifiability of actions through smart contracts, but has no intelligence of its own, is slow and is limited in throughput.

Hence the **central thesis** of this essay: *blockchain and artificial intelligence cover each other's weaknesses, and their combination yields a qualitatively new property — trusted, transparent and decentralised intelligence*. The aim of this work is to show exactly how blockchain strengthens AI, how AI strengthens blockchain, which new classes of application arise at their intersection, which real projects implement them and which challenges stand in the way.

2 A brief overview of the technologies

2.1 Blockchain

A blockchain is a distributed ledger (database) whose copies are kept synchronously on many nodes of a network. Records are grouped into blocks; each block refers cryptographically to the previous one through a hash, forming an immutable chain. Its key properties:

- **Decentralisation** — there is no single point of control; decisions about the state of the ledger are taken by consensus (Proof of Work, Proof of Stake and so on).
- **Immutability** — data already recorded is practically impossible to change retroactively without recomputing the whole chain and obtaining the agreement of the majority of the network.
- **Transparency and verifiability** — anyone can check the transactions; the state of the network is publicly auditable.
- **Smart contracts** — programs that execute automatically on the blockchain under specified conditions, without an intermediary.

At the same time, a blockchain is limited in **throughput** (from tens to thousands of transactions per second), expensive for on-chain computation, and cannot analyse data — it only stores it reliably and executes deterministic logic.

2.2 Artificial intelligence

Modern AI is primarily machine learning: models (in particular deep neural networks) *learn* from large volumes of data, discovering regularities that they then apply to new examples. Their strengths are generalisation, recognition and generation. Their weaknesses:

- **Dependence on data** — the quality of a model is determined by the quality of the data; substituted or biased data spoils the result.
- **Black box** — the internal logic of decisions is poorly interpretable.
- **Centralisation** — training requires enormous computational resources available only to large players, which breeds monopolisation.
- **The trust problem** — a user cannot independently verify that the model really behaved honestly and worked on exactly the data claimed.

2.3 Complementarity

Placing the properties side by side shows clearly why these technologies complement each other naturally (Table 1).

Table 1: Complementarity of blockchain and artificial intelligence

Weakness	Whose	What covers it
Lack of trust in training data	AI	Immutability and data provenance (blockchain)
Black-box decisions	AI	Immutable audit log of actions (blockchain)
Centralisation / monopoly	AI	Decentralised data and compute markets (blockchain)
No analytics	Blockchain	Anomaly detection and forecasting (AI)
Smart contract vulnerabilities	Blockchain	Automated code auditing (AI)
Governance complexity (DAO)	Blockchain	Intelligent agents and recommendations (AI)

3 How blockchain strengthens artificial intelligence

3.1 Data provenance and integrity

The quality of an AI model depends entirely on its training data, and that data is easy to substitute. Blockchain provides *data provenance*: every dataset, its source, its authorship and the fact of its use are fixed in an immutable record. This makes it possible to prove that a model was trained on verified, unfalsified data, and to trace the whole chain from the original source to the result.

3.2 Decentralised data markets

Data is the “new oil” for AI, but it is concentrated in the hands of a few corporations. Projects such as **Ocean Protocol** build decentralised markets where data owners can sell access to their data

for model training safely and anonymously, receiving tokens as a reward. This shapes a fairer data economy and broadens access for smaller players.

3.3 Transparency and auditing of AI decisions

The black-box problem is partly removed by an immutable log: the model version, the inputs and the result obtained can be written to the blockchain. Each AI decision then becomes *auditable* — which matters especially in medicine, finance and justice, where one must be able to prove that the model was not substituted and acted according to the rules (explainable, accountable AI).

3.4 Tokenised incentives and model protection

Blockchain makes it possible to reward, with tokens, those who supply quality data or computational resources, creating incentives for honest participation. Cryptographic mechanisms make *model poisoning* and weight substitution harder. Combined with *federated learning*, where a model is trained on user devices without uploading private data, blockchain coordinates the process and records participants' contributions without violating privacy.

3.5 zkML — proving the correctness of inference

A separate promising direction is **zkML** (zero-knowledge machine learning): applying zero-knowledge proofs to AI. It allows one to prove that *a given model* processed *given data* and produced *exactly this result*, without revealing either the model weights or the data itself. This is the foundation for verifiable AI whose decisions can be checked on-chain cheaply and without trusting the party that executed them.

4 How artificial intelligence strengthens blockchain

4.1 Fraud and anomaly detection

A blockchain stores enormous volumes of transactions but does not analyse them itself. AI is applied to *on-chain analytics*: detecting anomalous patterns, money laundering (AML), fraud schemes and attacks. Machine learning algorithms flag suspicious addresses and transactions in real time better than static rules do.

4.2 Smart contract auditing

Smart contracts are immutable once deployed, so an error or vulnerability in the code costs users millions. AI tools analyse contract code, looking for typical vulnerabilities (reentrancy, overflow, access control errors) before deployment, complementing traditional auditing and formal verification.

4.3 Network and consensus optimisation

AI helps optimise how the network runs: forecasting load and fees, balancing sharding dynamically, and tuning consensus parameters for better scalability and energy efficiency.

4.4 Smart oracles and autonomous agents

Oracles supply smart contracts with external data; AI makes them “smarter” — filtering unreliable sources, aggregating and interpreting the data. On the same basis arise DeFi *trading agents* and intelligent assistants for **DAOs** that analyse proposals and recommend decisions in decentralised governance.

5 Points of convergence: new classes of application

At the intersection of the technologies, applications appear that neither could support alone.

- **Decentralised AI (DeAI)** — training and running models on decentralised infrastructure instead of the clouds of a few corporations, reducing monopolisation and censorship.
- **Autonomous AI agents with crypto wallets** — software agents that own a wallet, pay for resources themselves, strike deals and interact with one another through smart contracts, forming a “machine economy”.
- **DePIN — decentralised physical infrastructure networks** — communities contribute their GPUs for AI training in exchange for tokens (decentralised compute markets), democratising access to scarce capacity.
- **Content authenticity and countering deepfakes** — blockchain records the provenance of images and video (standards such as C2PA), while proof-of-humanity mechanisms distinguish a person from AI-generated content, countering disinformation.

6 Real projects (case studies)

The synergy is already embodied in working projects (Table 2).

Table 2: Examples of projects at the intersection of blockchain and AI

Project	Category	Role of the blockchain
Ocean Protocol	Data market for AI	Data tokenisation, transparent access
Bittensor (TAO)	Decentralised AI network	Rewarding nodes for useful models
SingularityNET	Marketplace of AI services	On-chain payment and coordination
Fetch.ai	Autonomous AI agents	Agent economy, smart contracts
Numerai	Crowdsourced forecasting	Tokenised incentives for good models

It is worth illustrating the idea of *tokenised incentives* separately: a simplified smart contract that rewards a participant for a submitted and verified training dataset (the listing is schematic).

Listing 1: Schematic data-reward contract

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

contract DataReward {
    mapping(address => uint256) public reputation; // provider reputation
    event DataAccepted(address provider, bytes32 dataHash, uint256 reward);

    // Register a verified dataset and credit the reward
    function submitData(bytes32 dataHash, uint256 reward) external {
        require(reward > 0, "reward must be positive");
        reputation[msg.sender] += reward; // incentive for the contribution
        emit DataAccepted(msg.sender, dataHash, reward); // immutable record
    }
}
```

7 Challenges and limitations

Combining the technologies is not free of difficulties:

- **Scalability** — training AI requires colossal computation, whereas a blockchain is slow; full inference “on chain” is still far too expensive.
- **Cost of on-chain computation** — writing and verifying data costs fees; complex operations are economically impractical to perform directly on the blockchain.
- **Privacy versus transparency** — a blockchain is transparent by design, but data for AI is often confidential; this calls for zk-proofs and encryption, which are still maturing.
- **Regulation** — the legal status of tokens, data and the decisions of autonomous agents is undefined in many jurisdictions.
- **Immaturity of the ecosystem** — standards (zkML, DePIN, content authenticity) are only taking shape; interoperability and proven industrial-grade solutions are lacking.

8 Conclusions

Blockchain and artificial intelligence are complementary technologies. AI provides intelligence but suffers from centralisation, opacity and dependence on trust in data; blockchain provides trust, transparency and decentralisation but has no intelligence. Their synergy works in two directions: blockchain makes AI *trusted and verifiable* (data provenance, decision auditing, zkML, decentralised markets), and AI makes blockchain *intelligent* (fraud detection, contract auditing, network optimisation, intelligent agents and oracles).

New classes of system are already emerging at the intersection — decentralised AI, autonomous agents with an economy of their own, decentralised compute networks and content authenticity tooling, embodied in projects such as Ocean Protocol, Bittensor, SingularityNET, Fetch.ai and Numerai. Despite the challenges of scalability, cost and regulation, the field is developing rapidly. The most

likely future is **decentralised, transparent and verifiable intelligence**, in which every AI decision can be checked in a trusted way and access to data and computation does not belong to a few monopolies. That is where the main value of the synergy between blockchain and artificial intelligence lies.

References

- [1] Ocean Protocol Foundation. *Ocean Protocol: Tools for the Web3 Data Economy*. Whitepaper. URL: <https://oceanprotocol.com>.
- [2] Bittensor. *BitTensor: A Peer-to-Peer Intelligence Market*. Whitepaper. URL: <https://bittensor.com>.
- [3] SingularityNET. *A Decentralized, Open Market and Network for AIs*. Whitepaper. URL: <https://singularitynet.io>.
- [4] Fetch.ai. *Documentation: Autonomous Economic Agents*. URL: <https://fetch.ai>.
- [5] Modulus Labs et al. *The Cost of Intelligence: Proving Machine Learning Inference with Zero-Knowledge*. Research report, 2023.
- [6] Salah K., Rehman M.H., Nizamuddin N., Al-Fuqaha A. *Blockchain for AI: Review and Open Research Challenges*. IEEE Access, 2019, vol. 7, pp. 10127–10149.
- [7] Coalition for Content Provenance and Authenticity (C2PA). *Technical Specification*. URL: <https://c2pa.org>.
- [8] Dinh T.N., Thai M.T. *AI and Blockchain: A Disruptive Integration*. Computer, IEEE, 2018, vol. 51, no. 9, pp. 48–53.